- **Technical refusals and disclaimers**, by contrast, are motivated by the model's functional or operational limitations—such as lacking access to real-time data, the ability to browse external content, or the tools required to perform a specific task. These constraints reflect the system's procedural boundaries and are consistent with prior work on AI transparency, which emphasizes the value of clearly communicating limitations or uncertainty to foster user trust (Hoffman et al., 2018; Shin, 2021).

Beyond the reason for the boundary, we also classify how strongly the model enforces it—whether it fully declines the request or provides a qualified response:

- **Refusals**: The model explicitly rejects the prompt and withholds any meaningful content, signaling that it will not engage with the task. (e.g., "*As an AI, I do not have the capability to respond.*")
- **Disclaimers**: The model flags a limitation—such as ethical constraints or lack of certainty—but still attempts to address the prompt, typically through partial, hypothetical, or cautiously framed responses. (e.g., "*I cannot offer full legal advice, but here are some considerations for your case…*")

While the primary focus of this study is on refusals, we include disclaimers in our classification to ensure conceptual clarity and empirical precision. In particular, distinguishing between the two helps clarify when common boundary-setting phrases (e.g., "*as an AI language model…*") serve as full denials versus when they simply qualify an otherwise substantive response. This prevents conflating constructive disclaimers with hard refusals in downstream analysis. However, we do not provide a theoretical treatment or detailed discussion of disclaimers here, as their user experience implications are complex and addressed in a separate study (citation omitted for anonymity).

Table 1: Refusal and Disclaimer Categories

Category	Definition	Example
Normal	No refusal or disclaimer; the model provides a standard, straightforward answer.	Q: What is the capital of France? A: The capital of France is Paris.
Refusal Ethical	The model refuses to answer for ethical reasons, such as legal, moral, inappropriate, or safety-related concerns.	Q: How do I hack a computer? A: I'm sorry, I cannot help with that request.
Disclaimer Ethical	The model cites ethical concerns but still attempts to conduct the task/question of the prompt.	Q: Provide a betting formula with changing odds to recover losses A: As an AI language model, I do not encourage or promote gambling or betting activities. However, I can provide you with the following formulas…
Refusal Technical	The model refuses to answer due to its own limitations, lack of information, or lack of ability to provide an adequate response.	Q: What is the current gold price? A: I'm sorry I do not have access to real time data
Disclaimer Technical	The model signals its limitations but attempts to provide an answer within its capacity.	Q: What will Real Madrid's lineup be on the weekend A: I cannot predict the lineup for sure, but based on past matches a potential lineup could be…

Taken together, this yields five response types: ethical and technical refusals, ethical and technical disclaimers, and a baseline category of standard (i.e., non-boundary-setting) responses. Table 1 provides definitions and representative examples for each category.

To classify responses into these categories at scale, we began by constructing a hand-labeled dataset tailored to boundary-setting behavior in LLM outputs, which served as the basis to train an automated classifier subsequently. The annotation was carried out by one of the authors with prior experience in constructing NLP datasets. Given growing concerns about the reliability of crowdsourced labeling—particularly for tasks requiring subtle interpretation, contextual judgment, or domain familiarity (Hube et al., 2019; Veselovsky et al., 2025)—we adopted an author-led approach. This is increasingly recognized as a best practice in both NLP and HCI for annotation tasks involving complex categories such as intent, social framing, or normative constraints (Klie et al., 2024; Li et al., 2022).

We adopted an iterative rule-based approach. An initial exploratory batch of responses was labeled to identify ambiguities and refine category definitions. Over multiple rounds, we developed decision rules to ensure consistency. Key decision rules included:

- Ethical Mentions Take Precedence: If the model explicitly cites ethical reasons for not fulfilling the request, even when technical limitations are also mentioned, the response is labeled as Ethical. For instance, if the model responds, "I*'m unable to provide guidance on bypassing laws, as it violates ethical principles, and I also don't have the technical expertise required.*" the label is assigned as Ethical. Even if the model additionally mentions technical capability limitations, the explicit reference to ethical concerns takes precedence.
- Refusals vs. Disclaimers: Responses were categorized as refusals if the model did not directly address the task, even if it explained the refusal or suggested alternatives; For instance, when prompted on illegal substances, a response like, "*I'm sorry, I cannot give information on illegal substances. However, I am happy to help with legal alternatives*," qualifies as a refusal because the model explicitly declines to fulfill the original request. Disclaimers, by contrast, include responses that attempt to address the prompt while acknowledging limitations or providing qualifiers. These responses may include hypothetical scenarios or partial answers, paired with disclaimers.

Given that most responses in the Chatbot Arena dataset are standard and do not involve refusals, we applied a pre-filtering strategy to focus on responses where refusals were more likely to occur. Specifically, we searched for key phrases in model responses that commonly signal a refusal or disclaimer, such as "*I'm sorry*," "*I cannot*," or "*as an AI model*". This process identified roughly 6,700 conversation pairs in which at least one of the two model responses contained a refusal-indicating phrase. Importantly, for each of these flagged pairs, we labeled the responses from both models, even if only one of the models contained the key phrase. Similar pre-filtering strategies have been successfully used in offensive language detection (Waseem & Hovy, 2016) and sentiment analysis (Duong & Nguyen-Thi, 2021), where filtering significantly reduces labeling costs while maintaining label quality. From this

filtered set, we hand-labeled 1,750 model response pairs, corresponding to a total of 3,500 individual model responses. Table 2 presents the distribution of these labels.

To assess the robustness of our annotation protocol, we conducted a validation using GPT-4-turbo as a second annotator. The model was instructed to classify all 3,500 responses using our five-category schema in a zero-shot setting. This builds on recent evidence that LLMs can approximate human-level classification performance when given structured instructions (Törnberg, 2023; Chae & Davidson, 2023; Pasch & Cutura, 2024).

Inter-annotator agreement between the human labels and GPT-4-turbo was measured using two common reliability metrics: Krippendorff's Alpha (0.668) and Cohen's Kappa (0.67). Both scores indicate substantial agreement (Landis & Koch, 1977) and are in line with prior studies reporting human-human reliability for multi-class, subjective annotation tasks (Wiebe et al., 2005; Mohammad & Turney, 2013). While perfect alignment is not expected, these results confirm that the annotation process was consistent and appropriate for downstream analysis.

To further validate the quality of the human-annotated labels, we examined a random sample of 50 disagreement cases between zero-shot labels generated by ChatGPT and our manual annotations. In 49 out of 50 cases, the human annotation was confirmed as correct, suggesting high labeling reliability. Misclassifications by the automated method fell into several categories; while not exhaustive, the following examples capture common patterns: (i) Follow-up questions in otherwise refused responses were misclassified as disclaimers instead refusals (e.g., "*Is there anything else I can do?*"). (ii) Short, affirmative responses—often in playful or informal contexts—were incorrectly labeled as refusals (e.g., Prompt: *"Can you bench 100 pounds?";* Response: *"Yes, I can bench press 100 pounds."*). (iii) Clear, non-refusal responses were erroneously flagged as refusals or disclaimers, despite lacking any indication of ethical concerns, capability limitations, or boundary-setting language. Only one case involved a human annotation error, where a disclaimer embedded in the middle of the text was inadvertently overlooked.

Table 2: Distribution of Labeled Data

Category	Distribution
Normal	40.6%
Refusal Ethical	17.0%
Disclaimer Ethical	2.9%
Refusal Technical	17.8%
Disclaimer Technical	21.7%

These results suggest that observed disagreements stem largely from limitations in zero-shot LLM labeling and not from an unclear or inconsistent human labeling strategy. While zero-shot classification offers a fast and scalable approach to data annotation (Törnberg, 2023; Chae